F1 StratLab: An Open Multi-Agent System for Race Strategy Recommendation in Formula 1

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***Abstract*—Race strategy in Formula 1 decides Grands Prix on the basis of heterogeneous and uncertain information processed within seconds, yet despite the budgets that teams dedicate to internal decision-support systems, the public academic record on such systems is sparse and the few proprietary platforms in operation are closed. We present F1 StratLab, an open multi-agent system that issues lap-by-lap strategy recommendations from public data only. The system integrates four components over the 2023–2025 seasons: a feature-engineering pipeline that maps circuits onto four archetypes; a family of supervised predictors for lap time, tire degradation, pit-stop duration, undercut, overtake and safety-car probability, validated with a strict temporal split (2023–2024 train, 2025 test) to mitigate intra-cycle concept drift; a radio NLP pipeline combining Whisper transcription with fine-tuned sentiment, intent and entity models; and a retrieval-augmented generation subsystem grounded in the FIA Sporting Regulations. A three-layer orchestrator combines these components via rule-based routing, stochastic aggregation and structured language-model synthesis. Validation on three representative 2025 races shows that the system reproduces sound pit-wall reasoning; on two of these races it proposes alternatives that retrospective analysis judges better justified than the real team decisions. Code, weights and prompts are released under Apache 2.0, with a replay engine designed to accept real-time telemetry feeds and to be retrained on future regulatory cycles.**

***Index Terms*—Concept drift, Formula 1, large language model, machine learning, multi-agent system, natural language processing, race strategy, retrieval-augmented generation.**

I. Introduction

Formula 1 teams operate in one of the most data-rich environments of professional sport. A modern car streams several thousand telemetry channels at kilohertz rates and every race lap is shadowed by a strategy room that translates that signal into one of a handful of discrete actions: stay out, pit now, undercut a rival or overcut to defend track position. The economic and competitive stakes have pushed the leading constructors into multi-year alliances with specialised AI providers — Williams with Anthropic [1], Aston Martin with Cohere [2], McLaren with Dell [3] — yet the logic of the resulting decision-support tools is closed to the research community and to the wider F1 audience.

Producing a defensible race-strategy recommendation requires reconciling three heterogeneous planes of information. The first is *numeric and lap-by-lap*: tire degradation, lap-time evolution, pit-stop duration and the probability of an overtake or a safety-car intervention. The second is *linguistic*: team-radio communications between driver and pit wall, and race-control messages issued by the FIA, both of which condition the immediate decision space. The third is *regulatory*: every recommendation must comply with the FIA Sporting Regulations of the season under consideration, and a strategy engineer is expected to justify each call against a specific article. Combining these three planes within the latency budget of a live Grand Prix, while keeping every step auditable, is the core problem this paper addresses.

The academic record on F1 strategy treats these planes in isolation. Heilmeier et al. address race-strategy simulation under stochastic events with a Monte Carlo engine [4], [5]; Sasikumar et al. apply deep learning to pit-stop timing [6]; Urdhwareshe applies TabNet to season-aggregated race outcomes [7]; Cheng et al. use physics-informed neural networks for front-wing aerodynamics [8]; the Machine Learning at Cambridge Laboratory group has reported a confidential collaboration with an unnamed constructor, documented only at the announcement level [9]. None of these efforts integrates predictive modelling, radio understanding and regulatory grounding on a common, openly released stack over a complete regulatory cycle [10].

To close this integration gap we present *F1 StratLab*, an open multi-agent system that addresses the three planes end to end. Six specialised sub-agents (detailed in subsection III-C) wrap a family of supervised predictors trained on public FastF1 [11] and OpenF1 [12] data for the 2023 and 2024 seasons; a three-layer orchestrator (subsection III-D) routes their outputs, aggregates them through Monte Carlo sampling and synthesises a structured recommendation traceable back to the underlying prediction and to the regulatory article that supports it. The system is validated lap by lap on three representative 2025 races; on two of them, its recommendations propose alternatives that retrospective analysis judges better justified than the actual team decision. The present work extends two earlier projects by the author [13], [14], of which only the multimodal telemetry chat layer is reused as a Git submodule.

The contributions of this paper are as follows.

- A reproducible open corpus and a family of six supervised predictors for the core strategic outcomes (lap time, tire degradation, pit-stop duration, undercut, overtake and

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safety-car probability), with a strict temporal split (2023–2024 train, 2025 test) designed to expose and mitigate intra-cycle concept drift.
- A radio NLP pipeline that integrates Whisper transcription [15], fine-tuned RoBERTa sentiment [16], SetFit [17] and ModernBERT [18] intent classification and a BERT-large [19] entity recognizer, with a measured P95 latency of 59.4 ms on a single 8 GB consumer GPU.
- A retrieval-augmented generation subsystem indexed over the FIA Sporting Regulations of the 2023, 2024 and 2025 seasons with BGE-M3 dense embeddings, that attaches a citable regulatory article to each strategic recommendation.
- A three-layer orchestrator combining rule-based routing, Monte Carlo aggregation and structured language-model synthesis, designed for end-to-end traceability rather than opacity.
- End-to-end lap-by-lap validation on three reproducible 2025 case studies covering post-race analysis, an undercut window and a safety-car window, exercising the system on real races rather than synthetic scenarios.
- A complete open release under Apache 2.0: source code on GitHub, model weights and tokenisers on the Hugging Face Hub, prompts and orchestrator schemas in-repository, intended as a starting point for a community-maintained extension of the platform across future regulatory cycles.

The remainder of the paper is organised as follows. section II reviews the relevant work on concept drift, predictive models for motorsport, language understanding for radio communications, retrieval-augmented generation and multi-agent orchestration. section III details the system architecture and the three-layer orchestrator. section IV describes the datasets, the temporal validation protocol and the hardware footprint. section V reports per-agent and end-to-end results, including the three case studies. section VI discusses the known limitations and the boundaries of the public-data setting, and section VII closes with a community-oriented roadmap.

II. State of the Art

A. Prior work on AI for Formula 1 race strategy

The use of artificial intelligence to support race-strategy decisions in Formula 1 has been driven primarily by the constructors, through multi-year alliances with specialised technology providers [20]–[23]. Recent public announcements include Williams adopting Anthropic Claude as official "thinking partner" [1], Aston Martin integrating Cohere generative models [2] and McLaren publicising its analytics stack jointly with Dell [3]. These programmes are well funded and run at industrial scale, but they release only marketing material and broad architectural sketches; the predictive models, prompts, evaluation protocols and weights remain proprietary, which prevents external reproduction and academic comparison.

On the academic side, the published record is sparser and fragmented. Heilmeier et al. modelled pit-stop strategy with a Monte Carlo race simulator over Ergast data and showed that 500–1 000 trajectories suffice to compare candidate strategies under stochastic events [4], [5]. Sasikumar et al. applied recurrent and convolutional deep-learning architectures to pit-stop timing [6], and Urdhwareshe used TabNet to predict season-aggregated race outcomes from historical FastF1 data [7]. Cheng et al. applied physics-informed neural networks to front-wing aerodynamics [8], and the Machine Learning at Cambridge Laboratory group has reported a confidential collaboration with an unnamed F1 constructor [9]. Each of these contributions covers a single facet of the strategic loop. To the best of our knowledge, no published system integrates lap-by-lap prediction, team-radio understanding and regulatory grounding inside a single openly released stack over a complete regulatory cycle, nor does any of them address the structural concept drift introduced by the regulatory cycles that punctuate the sport, which we examine in the next subsection.

B. Concept drift in temporally non-stationary domains

Machine-learning models deployed on time-series data degrade when the joint distribution of inputs and target evolves between training and inference, a phenomenon catalogued under the umbrella of *concept drift* [10], [24]. Vela et al. document, across four sectors, that the temporal quality decay of production models is the norm rather than the exception [25]. Formula 1 is an extreme case of this regime: every multi-year regulatory cycle reshapes the aerodynamic and tire-supply rules, and a new season typically introduces a sudden drift that no within-season retraining can absorb. Within a cycle, gradual drift continues to operate through car development, tire-compound revisions and driver line-up changes. The methodologically legitimate evaluation protocol under these conditions is the temporal block cross-validation defended by Bergmeir and Benítez [26], in which training, validation and test sets respect strict chronological ordering. We adopt that protocol throughout the paper and quantify the residual drift within the 2023–2025 window of the ground-effect regulatory cycle, which ran from 2022 and closed with the 2025 season.

C. Predictive models for tabular and sequential signals

Race telemetry, once aggregated lap by lap, is a tabular problem with heterogeneous numerical and categorical features for which gradient-boosted decision trees remain the strongest off-the-shelf baseline [27]. The XGBoost [28] and LightGBM [29] implementations have become de facto standards in applied tabular machine learning, and modern histogram-based variants natively support quantile losses, which we exploit for pit-stop duration. Interpretability is provided by Shapley-value attributions [30], and calibrated probabilities, required before any downstream Monte Carlo aggregation, are obtained via Platt scaling [31]. For sequential signals such as tire degradation over a stint, recurrent networks [32] compete with temporal convolutional networks [33]; the latter offer larger receptive fields, stable training and explicit causal masking, which is critical when prediction at lap t must not see lap $t+1$. We quantify epistemic uncertainty by interpreting dropout as a Bayesian approximation [34], which yields the percentile bands that the orchestrator later consumes.

D. Language understanding for team radio and regulation retrieval

Team-radio communications and FIA race-control messages are the two linguistic streams that condition a strategic decision in real time. Robust automatic speech recognition on noisy radio is provided by the Whisper family of weakly supervised transformer encoders [15]. Downstream understanding builds on the bidirectional transformer encoders introduced by BERT [19] and the optimised pretraining of RoBERTa [16], which we fine-tune for radio sentiment over a lexicon-based VADER baseline [35]. For intent classification with very few labelled examples we rely on contrastive sentence embeddings popularised by Sentence-BERT [36] and on the few-shot SetFit framework [17], optionally backed by the recent ModernBERT encoder [18]. Regulatory grounding is implemented with retrieval-augmented generation [37] over the FIA Sporting Regulations, indexed with BGE-M3 multilingual dense embeddings [38] and queried through hierarchical navigable small-world graphs [39]. We favour token-level BIO tagging over span-extraction for the entity recogniser because our annotation scheme is clause-level and overlapping; the empirical comparison and its broader lesson are deferred to section VI.

E. Multi-agent orchestration with language models

The integration of heterogeneous predictors under a single controller has classical roots in mixtures of local experts [40] and modern echoes in the mixture-of-agents architecture of Wang et al. [41], where several large language models collaborate through structured rounds of reasoning. The ReAct prompting pattern [42] interleaves reasoning steps with tool invocations and is the basis of the sub-agents in our system. For the aggregation layer we follow the Monte Carlo race-simulation tradition of Heilmeier et al., who showed that a few hundred trajectories over discrete strategic candidates provide a robust expected-value comparison under stochastic events [4], [5]. These four strands — concept-drift methodology, predictive modelling for tabular and sequential signals, language understanding for safety-critical radio, and multi-agent orchestration — inform the architecture described in the next section, which integrates them into a single auditable Formula 1 strategy system.

III. Methodology and Architecture

A. System overview

F1 StratLab is organised in three logical layers: a data layer that ingests and harmonises every public source available for a race, a model layer that hosts six specialised sub-agents wrapping one or more supervised predictive models each, and an orchestration layer that fuses the sub-agent outputs into a single recommendation per lap. Figure 1 sketches the dataflow from the raw sources (FastF1, OpenF1, the FIA Sporting Regulations and the team-radio audio stream) through preprocessed lap states to the `StrategyRecommendation` Pydantic object delivered to the user.

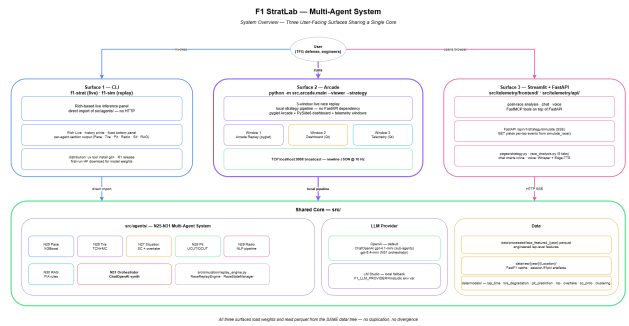

Figure 1. System architecture of F1 StratLab. Data sources (left) feed a unified per-lap state that drives six specialised sub-agents (centre); the orchestration layer (right) produces a single `StrategyRecommendation` per lap.

At each race lap the orchestrator receives a *lap state*, a typed snapshot that aggregates lap times, tire ages, gaps to rivals, weather, and any race-control message issued within the current window. The lap state is the single input contract for every sub-agent, which keeps the system loosely coupled and amenable to component-level retraining. Sub-agent outputs are likewise typed Pydantic schemas, so the orchestrator composes them deterministically. The result is a system in which every strategic recommendation is traceable back to the prediction, the regulatory article and the radio alert that justify it, a property we exploit in the three-layer orchestrator described below.

B. Data ingestion, cleaning and feature engineering

We consolidate the 2023, 2024 and 2025 race seasons by merging the FastF1 telemetry API [11], the OpenF1 REST API [12] and the FIA race-control message stream into a unified per-lap table of 22 106, 23 256 and 22 760 laps, respectively. Race-control messages are parsed deterministically by a rule-based extractor, with 100 % coverage on flag, safety-car and DRS events and 93 % on the free-form notes, before they are attached to the lap they affect. Outliers from safety cars, virtual safety cars and red-flag restarts are flagged rather than removed, so that downstream models can decide whether to drop or condition on the affected laps.

From the consolidated table we build a 53-column featurized dataset, including fuel-corrected lap times, intra-session relative normalisation (z-scores against the rolling stint baseline) and short-horizon deltas at 1-, 3- and 5-lap windows; each agent consumes a model-specific subset of these columns. These features are deliberately classical rather than learned, in line with the empirical finding that domain knowledge dominates over deep representations on lap-level tabular data of this size; we return to that evidence in section V.

We cluster the 24 calendar circuits into $K{=}4$ archetypes with k-means [43], [44] over a 14-dimensional descriptor (silhouette 0.20 at $K{=}4$ versus 0.17 at $K{=}5$). The four archetypes correspond to *strategic chaos* circuits (high overtaking, mixed pace), *long high-speed* circuits, *warm conservative* circuits and a small group of *high-stress outliers* (Monaco, Singapore, Las Vegas). The cluster identifier is consumed by the pace, tire and race-situation agents, but is excluded from the pit-strategy models: its statistical contribution did not survive a permutation-importance test on the 2024 validation fold and would have introduced spurious cross-circuit transfer.

C. Specialised sub-agents

Pace agent. An XGBoost [28] regressor trained on the lap-time delta against the previous lap, rather than on the absolute time, removes the bulk of inter-driver and fuel-related variance. Bootstrap resampling with 200 draws and a 2 % Gaussian perturbation supplies the P10/P90 bands that the orchestrator's Monte Carlo layer later consumes as the lap-time distribution.

Tire agent. A temporal convolutional network [33] with $\sim$100 k parameters and causal masking predicts the laps-to-cliff over per-stint sequences of up to 42 features. Causal convolutions are non-negotiable here because the model is queried lap by lap during a live race, and any leakage from future laps would invalidate the prediction. We train one global model on 2023–2024 and, guided by per-compound holdout evidence, fine-tune a single variant for compound C2; fine-tuning was ruled out for C1, C3, C4 and C5 because their holdout performance did not improve (neutral for C1 and C3, regressed for C4 and C5). Epistemic uncertainty is quantified by 50 Monte Carlo dropout [34] forward passes that yield the P10/P50/P90 cliff window.

Race-situation agent. Two LightGBM [29] classifiers, one for overtake probability over the next three laps and one for safety-car probability over the same horizon, are calibrated with Platt scaling [31] on the 2024 validation fold so that the Monte Carlo aggregator receives valid probabilities. The overtake classifier is genuinely discriminative; the safety-car classifier acts as a soft contextual prior rather than a fine-grained predictor, and the orchestrator weighs it accordingly.

Pit-strategy agent. Activated conditionally when the tire agent reports `PIT_SOON` or when the radio agent flags a `PROBLEM`/`WARNING`. A histogram-based gradient-boosted quantile regressor predicts pit-stop duration as a P05/P50/P95 triple, and a LightGBM binary classifier with Platt calibration predicts the success of an undercut against a specific rival. The duration is decomposed into a fixed circuit-traversal component and a variable physical-stop component, which we found necessary to keep error stable across pit lanes of very different geometry.

Radio agent. An NLP-first pipeline [15]–[19] chains Whisper Turbo automatic speech recognition, a fine-tuned RoBERTa sentiment classifier, a SetFit and ModernBERT intent classifier, a BERT-large BIO entity recogniser and a rule-based race-control parser. We deliberately make the alert-extraction step deterministic and execute it *before* the orchestrator's LLM synthesis stage. A ReAct loop would let the LLM choose whether to extract a safety alert; for a system that may recommend a pit under a yellow flag, that choice cannot be left to a model.

Regulation agent. Conditionally activated when the safety-car probability exceeds 0.30, when a pit decision is on the table or when the radio reports a penalty. We index the FIA Sporting Regulations for the 2023, 2024 and 2025 seasons as 2 279 chunks of 512 characters with 64-character overlap, encode each chunk with BGE-M3 dense embeddings [38] into a 1 024-dimensional space and serve them through a Qdrant index with hierarchical navigable small-world graphs [39]. Retrieved articles are returned verbatim to the orchestrator and quoted in the final recommendation, which keeps every regulatory claim traceable to a specific paragraph of the rulebook [37].

Table I consolidates each sub-agent's model, predicted outcome, uncertainty mechanism and output schema.

Table I
SPECIALISED SUB-AGENTS OF F1 STRATLAB.

Agent	Model	Uncertainty	Output schema
Pace	XGBoost on lap-time delta	Bootstrap, 200 draws	`lap_time_pred`, P10/P90 bands
Tire	Causal TCN, global plus C2 fine-tune	MC Dropout, $N{=}50$ passes	`laps_to_cliff` P10/P50/P90, warning
Race situation	LightGBM ×2 (overtake, SC)	Platt scaling on 2024 fold	overtake and safety-car probabilities
Pit strategy	HistGBT quantile plus LightGBM	Pinball quantile plus Platt	duration P05/P50/P95, undercut probability
Radio	Whisper, RoBERTa, SetFit, BERT-BIO, rules	Deterministic NLP-first extraction	`alerts[]`, sentiment, intent, NER
Regulation	BGE-M3 plus Qdrant HNSW	Multi-query fallback retrieval	retrieved `articles[]`, top-$k{=}5$ chunks

D. Three-layer orchestrator

The orchestrator is implemented as a manual three-layer supervisor (Figure 2): a rule-based mixture-of-experts router that decides which conditional agents to invoke and runs them, a Monte Carlo aggregator that scores four discrete strategic actions under the joint uncertainty exposed by the sub-agents, and a language model that synthesises the result into a structured recommendation. We chose a manual router over a learned gating function because no labelled corpus of optimal sub-agent combinations exists for this problem and because the per-lap latency target rules out a conversational router.

Layer 1: routing and invocation. Four sub-agents are invoked at every lap (pace, tire, race situation, radio); two are conditional and gated by four `if-else` rules

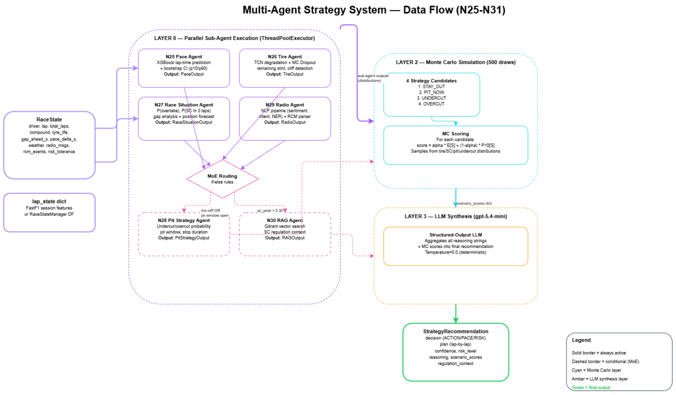

Figure 2. Three-layer orchestrator pipeline. The MoE-style router (left) decides which sub-agents to invoke and runs them in the appropriate concurrency mode; a 500-trajectory Monte Carlo aggregator (centre) scores four discrete strategic actions; a language model (right) synthesises the structured `StrategyRecommendation`.

with a cascade. Pit strategy is activated when the tire agent signals `PIT_SOON` or when the radio agent flags a `PROBLEM`/`WARNING`; regulation retrieval is activated when the three-lap safety-car probability exceeds 0.30, when the radio flags a `PENALTY`/`WARNING`, or whenever pit strategy is active so every pit decision is checked against the rulebook. Thread-safe sub-agents (XGBoost pace, LightGBM race situation) run in parallel through a thread pool; the PyTorch tire model and the MLX-backed radio pipeline are serialised to avoid conflicts in their respective inference runtimes.

Layer 2: Monte Carlo aggregation. The aggregator draws 500 independent trajectories per lap, with fixed seed, from five distributions that materialise the joint uncertainty of the activated sub-agents: a normal distribution for lap time [45] parameterised by the pace agent's P10/P90, a triangular for the laps-to-cliff over the tire agent's P10/P50/P90 [46], Bernoulli variables for safety-car presence and undercut success, and a triangular for pit-stop duration. Each sample is evaluated under four strategic candidates — `STAY_OUT`, `PIT_NOW`, `UNDERCUT` and `OVERCUT` — through a five-lap window function with the five calibrated constants of Heilmeier et al. [4]. Each strategy is finally scored as $\alpha\, E[S] + (1{-}\alpha)\, P_{10}[S]$, where $E[S]$ is the expected score, $P_{10}[S]$ the 10th percentile of the score distribution (a worst-case lower bound), and $\alpha \in [0, 1]$ (default 0.5) trades expected value against worst-case protection, following the conservative mixed-strategy principle of Liu et al. and Wang et al. [41], [47].

Layer 3: language-model synthesis. The final layer is a single call to `gpt-5.4-mini` [48] at temperature zero with structured output enforced against a 14-field Pydantic schema (`StrategyRecommendation`). The prompt receives, verbatim, the numeric outputs of every activated sub-agent, the Monte Carlo summary table, any retrieved regulatory articles, and six explicit guardrails that veto infeasible decisions: a five-lap pit lockout at race start (unless a safety car is deployed or radio reports confirm damage), a three-lap lockout at race end (unless a tire cliff is imminent), a restriction of the `REACTIVE_SC` action to safety cars actually on track, a minimum-stint-length floor per compound (8, 12 and 15 laps for soft, medium and hard), a compatibility check between the chosen compound and the remaining laps, and a one-level discount on the race-situation threat during the first three laps to absorb the post-start chaos. The schema enforces a strict separation between the discrete action, the pit plan, the pilot directives and the contingency list, so that each block can be validated independently against the previous two layers. Every recommendation is therefore traceable from the action it delivers back to the model that produced its quantitative basis, the article that grounded its regulatory check and the radio alert that triggered the conditional path.

IV. Experimental Setup

A. Datasets and corpora

The system is evaluated against three public corpora drawn from the 2023, 2024 and 2025 Formula 1 seasons. The telemetry corpus combines FastF1 [11] and OpenF1 [12] and provides 22 106, 23 256 and 22 760 lap records for each of those seasons respectively, with the associated tire compounds, sector times, gaps and race-control events. The team-radio corpus contains 530 hand-transcribed and hand-labelled messages, with an additional sub-corpus of 28 messages and 76 race-control items verified on the 2025 Bahrain Grand Prix as an integration ground truth. The regulatory corpus is the union of the FIA Sporting Regulations for 2023, 2024 and 2025, segmented into 2 279 chunks of 512 characters with a 64-character overlap.

B. Validation protocol

We adopt a strictly temporal hold-out protocol throughout the paper. Hyperparameter search is run with the 2023 season as the inner training fold and the 2024 season as the inner validation fold, so that no future race is ever visible during model selection. Once hyperparameters are fixed, each production model is retrained on the union of 2023 and 2024 and evaluated on the 2025 season, which is held out as the unseen test set throughout the paper. Random cross-validation is deliberately avoided because it would conflate the intra-cycle concept drift with random sampling noise and break the assumption of independent and identically distributed sampling. Probabilistic classifiers (overtake, safety car, undercut) are calibrated with Platt scaling [31] on the 2024 fold so that the Monte Carlo aggregator of the orchestrator receives valid probabilities. We report mean absolute error and R^2 for regressors, AUC-PR and AUC-ROC for binary classifiers, pinball loss at P05, P50 and P95 for the quantile regressor, macro F_1 for the NLP classifiers, and P95 end-to-end latency for the integration pipeline.

C. Model selection and ablations

The pace agent compares XGBoost, random forest and LightGBM over the lap-time delta target rather than the absolute lap time; the delta formulation removes the bulk of inter-driver and fuel-related variance and is the design that finally enters production, with XGBoost dominating the validation fold.

The tire-degradation TCN is the most extensively ablated component of the system. Five versions are compared, isolating the contribution of HuberLoss with δ=1 s, which down-weights safety-car and red-flag outliers relative to a mean-squared loss; per-lap stint sampling, which inflates the 1 081-sequence corpus to 20 363 prefix samples without collecting new data; and the choice of optimiser schedule, where stochastic-gradient-descent warm restarts proved harmful in Phase 2 by interrupting fine-tuning mid-cycle, so we reverted to a monotonic cosine schedule. A documented negative result is the failure of reversible instance normalisation in version 5: it modestly improved MAE but collapsed R^2, evidence that the dominant residual error is long-stint extrapolation rather than inter-season shift. Per-compound fine-tuning is then routed by holdout evidence: only compound C2 improves, while C1, C3, C4 and C5 regress, so all four fall back to the global model.

For the race-situation agent, the overtake classifier is tuned with 100 Bayesian trials maximising AUC-PR on the 2024 fold and the F_1-optimal decision threshold is set at 0.80. The safety-car classifier was attempted on prediction horizons of three, five and seven laps; we keep the three-lap variant because the longer horizons added no signal beyond a flat prior. A TCN variant of the overtake model (N12B) was trained and discarded: its AUC-PR of 0.099 against the LightGBM baseline of 0.549 illustrates that classical domain features dominate over deep sequence representations on the scale of this corpus.

The pit-strategy duration model adopts histogram-based gradient boosting with a pinball loss in preference to a mean-squared regressor, because the downstream agent needs the probability that the duration falls within a pit window rather than its expected value. The duration is decomposed into a fixed circuit-traversal component and a variable physical-stop component, which we found necessary to keep the error stable across pit lanes of very different geometry. A permutation-importance probe flagged the `TyreAge` feature with a strongly negative score, which we interpret as a leakage signal from 2023–2024 patterns absent in 2025 and discuss further in section VI. The undercut classifier is calibrated with Platt scaling, and its decision threshold is tuned at 0.522 by the same Bayesian search rather than left at the default of 0.5.

For radio understanding, a VADER lexical baseline [35] is rejected because of the gap between its sentiment distribution and the manually labelled corpus (33.1 % neutral predictions versus 71.5 % in our labels); a fine-tuned RoBERTa is the production choice. For intent classification, DeBERTa-v3 with 304 M parameters failed to converge on the 370-example training corpus and was replaced by a SetFit head over the ModernBERT encoder, which is the production configuration. For named-entity recognition over clause-level domain tags, a BERT-large encoder with BIO tagging reaches F_1 0.42, while a GLiNER baseline reaches only 0.07 once fine-tuned on the same 319 examples (and 0.016 zero-shot): paradigm fit dominates over model size when annotations are clause-level and can overlap.

For regulatory retrieval, BGE-M3 with 1 024-dimensional dense embeddings is the production encoder; smaller English-only alternatives were evaluated but performed worse on the multilingual boilerplate of the FIA corpus.

Taken together, these ablations reveal a consistent lesson: classical domain features and paradigm-fit annotations outperform raw model scale on the tabular and sparse-label problems characteristic of this corpus.

D. Hardware, software and reproducibility

All experiments run on a single workstation with an NVIDIA GeForce RTX 5070 Laptop GPU (8 GB GDDR7) and an AMD Ryzen AI 9 365 CPU, which is the same envelope under which the system is intended to serve in production. The software stack is Python 3.11, PyTorch with CUDA 12.8 on the Blackwell architecture, XGBoost, LightGBM and HistGradientBoosting via scikit-learn, the Hugging Face `transformers` library for the BERT family and Whisper, Qdrant as the vector store and LangGraph for the orchestrator plumbing. The orchestrator's synthesis layer uses the OpenAI API, with `gpt-4.1-mini` [49] for the sub-agents and `gpt-5.4-mini` [48] for the supervisor and the conversational backend. Reproducibility is supported by fixed random seeds, a deterministic replay engine, pinned dependency versions, model weights and preprocessing artefacts distributed through the Hugging Face Hub, and source code released under the Apache 2.0 licence.

V. Results Analysis

A. Supervised predictors

Table II consolidates the 2025 holdout metrics for the supervised predictive layer. All results are obtained under the strictly temporal protocol of section IV: no 2025 data enters any training or validation split.

Table II
SUPERVISED PREDICTOR RESULTS ON THE 2025 HOLDOUT. BASELINE: NAÏVE PERSISTENCE (LAP TIME), BASE RATE ON TEST SET (CLASSIFIERS), TEAM×CIRCUIT MEDIAN (PIT DURATION).

Component	**Metric**	**Value**	**Baseline**
Lap-time delta	MAE (s)	0.410	0.408
(XGBoost)	R^2	0.9947	—
Tire global	MAE (s)	0.708	—
(TCN v4)	R^2	0.605	—
Tire C2 fine-tune	MAE (s)	0.550	0.708
Overtake	AUC-PR	0.549	0.076
(LightGBM)	AUC-ROC	0.876	0.500
Safety car	AUC-PR	0.072	0.043
(LightGBM)	Lift	1.67×	1.00
Pit duration	P50 MAE (s)	0.487	0.555
(HistGBT quant.)	Pinball P05/P95	0.038 / 0.110	—
Undercut	AUC-ROC	0.771	0.500
(LightGBM)	AUC-PR	0.674	0.345
	Lift	1.95×	1.00

The lap-time delta formulation removes the bulk of inter-driver and fuel-related variance; the R^2 of 0.9947 on the 2025 holdout confirms that the residual error reflects race-specific tactical noise rather than systematic calibration drift. The naïve persistence baseline posts a nearly identical MAE of 0.408 s, which underscores that the regressor's value lies not in marginal absolute accuracy but in supplying calibrated P10/P90 bands that the Monte Carlo aggregator propagates across 500 trajectories, a capability that no persistence heuristic can provide. The global tire TCN reaches MAE 0.708 s; the C2 fine-tune variant reduces this to 0.550 s. Fine-tuning was ruled out for C1, C3, C4 and C5 after the 2024 holdout showed neutral or regressive performance relative to the global model, because the 1 081-sequence corpus provides too few full-stint samples per compound for reliable specialisation.

The overtake classifier exceeds the base-rate AUC-PR by a factor of 7.2, confirming that the engineered `pace_delta_rolling3` and `gap_trend` features carry genuine discriminative signal at the three-lap horizon (Figure 3) — a result the causal TCN variant (N12B, AUC-PR 0.099) failed to replicate, as reported in subsection IV-C.

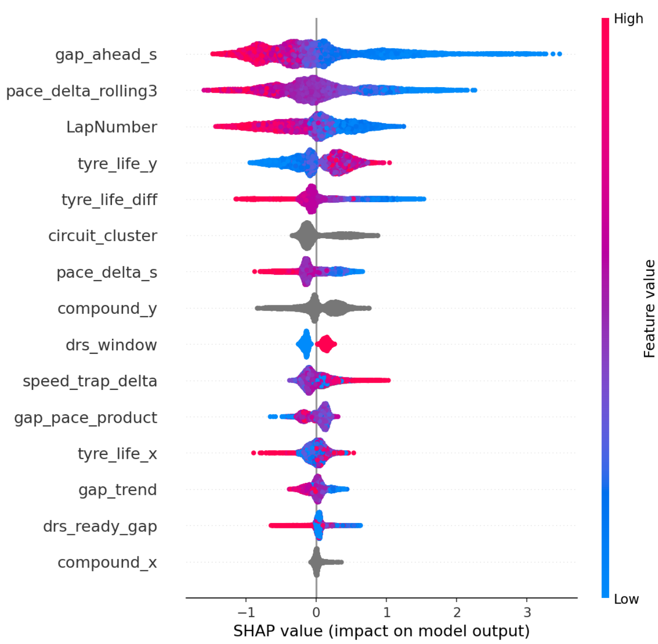

Figure 3. SHAP summary for the overtake classifier on the 2025 holdout. The pace-delta rolling mean and the gap-trend feature dominate the prediction surface, with sign and magnitude consistent with on-track overtaking dynamics.

The safety-car classifier posts AUC-PR 0.072 with lift 1.67× at the F2-optimal threshold of 0.234; it is treated as a contextual risk prior inside the Monte Carlo aggregator rather than a binary trigger, consistent with the handling of rare-event classifiers in sequential decision systems [10]. A permutation-importance probe on the 2025 pit-duration fold flagged `TyreAge` with a strongly negative score of -0.848 s in MAE, larger in absolute value than the model's headline MAE of 0.487 s. We interpret the sign as feature-level concept drift between the 2023–2024 Pirelli tyre construction and the 2025 revision; the model retains the feature for within-season generalisation, though its cross-season predictive value is limited by the compound construction change.

B. NLP pipeline and regulatory retrieval

Table III reports per-task NLP metrics and end-to-end pipeline latency measured over the test corpus on the RTX 5070 GPU.

Table III
NLP PIPELINE AND REGULATORY RETRIEVAL RESULTS. PIPELINE LATENCY MEASURED OVER 100 MESSAGES.

Task / Component	**Metric**	**Value**
Sentiment (RoBERTa fine-tuned)	Accuracy	0.84
	Macro F_1	0.75
Intent (SetFit+ModernBERT)	Accuracy	0.61
	Macro F_1	0.53
NER (BERT-large BIO)	Token F_1	0.415
Pipeline E2E (Whisper→NER)	Mean GPU (ms)	47.8
	P95 GPU (ms)	59.4
Regulation retrieval (BGE-M3 + Qdrant)	Content P@5	0.800
	MRR	0.235
	P95 (ms)	243.7

The WARNING intent class collapses to F_1 zero on the test fold (five test examples); the production pipeline routes it conservatively to `PROBLEM` handling to avoid false negatives on safety-critical radio events. The BERT-large BIO recogniser reaches token F_1 0.415 on nine clause-level entity types, dominating all evaluated alternatives: GLiNER reaches 0.07 once fine-tuned on the same corpus and 0.016 zero-shot, confirming that BIO sequence tagging matches the overlapping clause-level annotation structure of F1 radio more faithfully than span-extraction approaches at this corpus size. The residual gap to higher F_1 reflects the corpus envelope: 530 messages hand-transcribed and hand-annotated by a single labeller against nine entity types, with no inter-annotator agreement protocol; we estimate that crossing $F_1 \approx 0.60$ would require expanding the corpus to roughly 2 000 messages with at least two annotators.

The end-to-end pipeline operates at a GPU P95 of 59.4 ms (mean 47.8 ms), well within the 500 ms operational budget for a radio message arriving between laps; CPU latency is not measured because production deployment assumes a GPU host. The regulatory retrieval achieves Content P@5 of 0.800, meaning the semantically correct article appears within the top five results in four of every five queries. The gap to exact-match P@5 (0.200) reflects that compound regulatory questions typically require synthesising two or three retrieved passages rather than a single top-ranked result.

C. System integration and case studies

Integration smoke tests. A full-session smoke test on the Melbourne 2025 Grand Prix confirms complete orchestration without assertion failures across all race laps. The Bahrain 2025 integration test processes 28 team-radio messages and 76 race-control items; the radio agent correctly raises a PROBLEM alert at lap 4 from a Hamilton transmission and activates the pit-strategy conditional path as intended. The radio pipeline benchmark of Table III confirms that the dominant NLP latency cost (BERT-large named-entity recognition at P95 59.4 ms) stays well inside the orchestrator's per-lap window; the heavier regulation-retrieval call (P95 243.7 ms) only fires on laps where a safety-car event, a penalty or an active pit decision is signalled, covering roughly 11–18 % of laps per race.

Case study 1: Australia 2025 (Streamlit, post-race analysis). The Streamlit MCP interface is exercised over the Australian Grand Prix with Norris and Piastri as the monitored pair across laps 13–25. Strategy summaries and lap-time comparisons are delivered through the `get_lap_times` tool call over Server-Sent Events. Across all monitored laps the orchestrator recommendations align with the McLaren pit-wall decisions taken in the race (Figure 4). This provides a qualitative confirmation that the system introduces no systematic recommendation bias on routine strategic situations and that the full Streamlit–FastMCP–orchestrator stack completes all tool calls and SSE deliveries without failures.

Figure 4. Streamlit chat interface during the Australian Grand Prix post-race analysis (case study 1). The MCP tool call `get_lap_times` streams lap-time deltas between Norris and Piastri across laps 13–25 via Server-Sent Events.

Case study 2: Hungary 2025 (CLI, undercut window). Leclerc on Ferrari (hard compound) faces Piastri on McLaren across laps 17–19 of the Hungarian Grand Prix, replayed as:

```
f1-sim Budapest LEC Ferrari
  --year 2025 --laps 17-19
  --rival PIA --provider openai
```

The orchestrator returns PIT_NOW at confidence 0.94 at lap 17 and 0.97 at lap 18, driven by the undercut classifier flagging a viable gap against Piastri and the tire agent issuing a cliff warning. At lap 19 the scoring shifts to OVERCUT at 0.73, because the undercut window has partially closed. Ferrari pitted at lap 22 on the hard compound, losing the undercut opportunity and finishing behind Piastri (Figure 5). Gap-evolution analysis confirms that a lap-17 or lap-18 stop would have achieved track position.

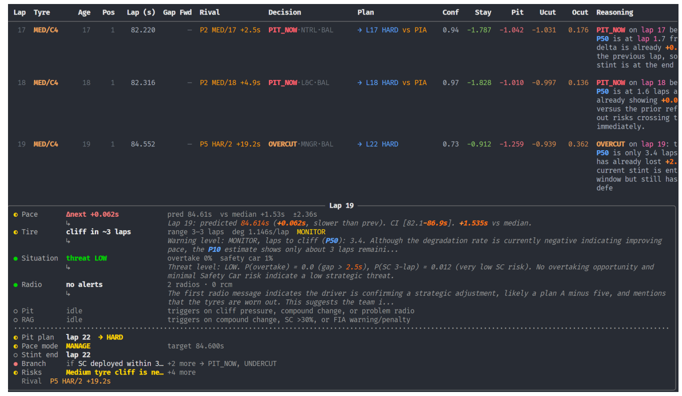

Figure 5. CLI simulator output at lap 19 of the Hungarian Grand Prix (case study 2). The orchestrator shifts to OVERCUT at confidence 0.73 after the undercut window against Piastri has partially closed in the preceding laps.

Case study 3: Qatar 2025 (Arcade, safety-car window). A safety car at lap 7 of the Qatar Grand Prix opens a pit window for Piastri (McLaren) against Verstappen, replayed as:

```
f1-arcade --viewer --year 2025
  --round 23 --driver PIA
  --team McLaren --driver2 VER
  --strategy
```

Without the `RCMContextResolver` module, the orchestrator returns STAY_OUT at 92%, reproducing the McLaren pit-wall decision. The module post-processes the race-situation output against the active race-control message stream, injects a `sc_currently_active` boolean into `RaceSituationOutput` and, when that field is set, disables the minimum-stint guardrail in the pit-strategy agent. With the module active, the orchestrator returns PIT_NOW at 97%, citing the Pirelli 25-lap tyre-usage mandate at Lusail (Article 36.3) retrieved by the regulation agent (Figure 6). McLaren stayed out, ceding track position to rivals who pitted under the safety car; retrospective analysis by multiple commentators identified this as the team's worst strategic error of the season, and the corrected system recommendation aligns with the post-race expert consensus.

VI. Discussion and Known Limitations

The most consistent empirical lesson across all three modelling domains is that paradigm fit and domain-specific feature engineering outperform raw model scale under the data-volume constraints of this corpus. In the tabular domain, the LightGBM overtake classifier with engineered pace-delta and gap-trend features reaches AUC-PR 0.549; a causal TCN over the same sequences reaches 0.099. In intent classification, SetFit over ModernBERT converges on 370 labelled examples where DeBERTa-v3 (304 M parameters) fails to converge. In named-entity recognition, BERT-large BIO tagging reaches token F_1 0.415 on clause-level radio entities where the GLiNER

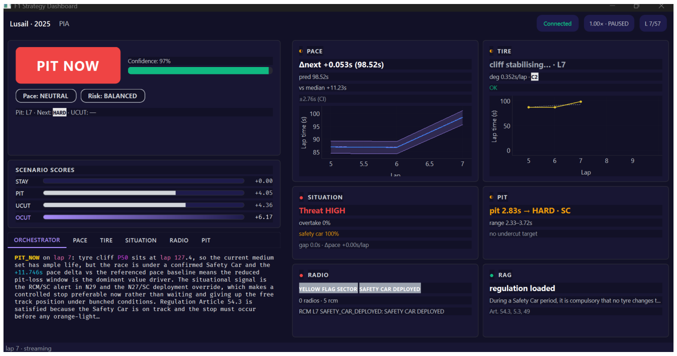

Figure 6. Arcade strategy panel at lap 7 of the Qatar Grand Prix under safety car (case study 3). With the `RCMContextResolver` module active, the orchestrator returns PIT_NOW at 97% citing the Pirelli 25-lap mandate at Lusail (Article 36.3).

span-extraction model reaches 0.07 fine-tuned and 0.016 zero-shot. This pattern is consistent with the broader finding that encoding known domain structure into the feature or label space outperforms learning it end-to-end when annotated data is scarce, a regime that characterises all three components of this corpus.

The public-data constraint sets a hard ceiling on tire-degradation modelling in particular. Internal thermocouple readings and pressure sensors are not exposed by FastF1 or OpenF1; the global TCN's MAE of 0.708 s likely reflects this informational gap rather than an architectural shortcoming. The per-compound fine-tuning gain on C2 is bounded by the number of full-stint sequences available per compound, not by model capacity. The negative permutation importance of `TyreAge` on the 2025 pit-duration holdout provides direct evidence that Pirelli compound construction changes between seasons introduce feature-level concept drift that temporal cross-validation alone cannot eliminate [10]: the feature encodes tyre behaviour that differed qualitatively in 2025 relative to the 2023–2024 baseline on which it was calibrated.

The safety-car AUC-PR of 0.072 deserves contextualisation before it is read as a limitation. The 2025 test-set base rate is 0.043; any discriminative signal above that floor is operationally useful, and the classifier delivers lift 1.67× at the F2-optimal threshold of 0.234, a threshold deliberately biased toward recall to avoid missing a real safety-car window. Achieving high precision under these conditions would require signals outside the scope of public data, such as live marshalling-point feeds or internal FIA communications. Under public-data constraints the classifier is correctly deployed as a continuous risk weight in the Monte Carlo scoring function rather than as a binary trigger.

The named-entity recogniser is the NLP pipeline's weakest component; its token F_1 of 0.415 reflects two structural limitations of the labelled corpus rather than the model architecture. The 530 Whisper-transcribed radio messages used for sentiment and intent training collapse to 399 messages once filtered for the dedicated NER annotation pass, and that NER subset was annotated solely by the author without inter-annotator agreement measurement, which introduces systematic inconsistency on ambiguous tokens such as compound names used as adjectives and tyre-age references without explicit compound context. Additionally, the WARNING intent class collapses to F_1 zero on the test fold because only five test examples are available; the production pipeline circumvents this by routing WARNING conservatively to `PROBLEM` handling, but the class remains undertrained. Scaling the corpus to approximately 2,000 examples with a multi-annotator protocol is the immediate path to meaningful improvement on both components.

End-to-end evaluation is necessarily qualitative at the recommendation level: no ground truth of optimal pit-wall decisions exists for the three case-study races, and the comparison with actual team behaviour is retrospective rather than prospective. The system's recommendations are assessed against observable race outcomes and against the pit-wall logic that race engineers publicly disclosed post-race, a standard of evidence appropriate to open-data academic work but not equivalent to a controlled A/B test against professional strategists. A season-level evaluation protocol — recording the orchestrator recommendation and the ground-truth team decision on every strategically relevant lap across all 24 races and computing concordance over the four discrete actions — is the next concrete step in establishing quantitative external validity.

The ground-effect regulatory cycle on which the system is calibrated ran from 2022 and closed with the 2025 season, which makes the 2025 holdout the final state of that cycle and the 2026 rule change its planned obsolescence boundary. The new regulations introduce a power-unit architecture with a 50% electrical power share, a revised aerodynamic concept and the elimination of the MGU-H, all of which will shift tyre-degradation dynamics and circuit archetype structure in ways the present models cannot anticipate. The system is explicitly built for this transition rather than against it: model retraining requires re-running the notebook pipeline on updated data, the circuit clustering is re-derived from new telemetry, and the regulation index is refreshed by pointing the download script at the 2026 FIA Sporting Regulations PDF. The modular architecture described in section III makes this a planned maintenance cycle rather than a research contribution.

VII. Conclusion and Future Work

This paper presents F1 StratLab, an open multi-agent system that issues lap-by-lap race strategy recommendations from public Formula 1 data by integrating tabular predictive models, a radio natural-language processing pipeline, a retrieval-augmented regulation index, and a three-layer orchestrator in a single reproducible stack. The supervised predictors achieve competitive accuracy on the strict 2025 temporal holdout: the lap-time regressor reaches MAE 0.410 s with R^2 0.9947, the overtake classifier reaches AUC-PR 0.549 against a base-rate of 0.076 (7.2× lift), and the undercut predictor achieves AUC-ROC 0.771. The radio pipeline combines Whisper ASR, fine-tuned RoBERTa sentiment (accuracy 0.84), SetFit intent classification, and BERT-large BIO entity recognition at a P95 GPU latency of 59.4 ms. End-to-end validation on three 2025 Grands Prix demonstrates that the system reproduces sound pit-wall reasoning on routine calls and surfaces better-justified alternatives on the non-routine ones; the Qatar safety-car case study in particular shows that the `RCMContextResolver` module surfaces the pit recommendation that McLaren's pit wall later acknowledged they had missed.

Beyond the motorsport domain, the system offers three methodological lessons. First, temporal block cross-validation is the only valid evaluation protocol in domains punctuated by regulatory cycles; random cross-validation inflates reported performance by conflating concept drift with sampling noise, an effect directly observable in the negative permutation importance of `TyreAge` on the 2025 holdout. Second, paradigm fit outperforms raw model scale under sparse-label constraints: BIO sequence tagging reached F_1 0.415 on clause-level radio entities where a GLiNER span-extraction model reached 0.02 zero-shot, and SetFit converged on 370 intent examples where DeBERTa-v3 at 304 M parameters did not. Third, conditional agent routing is a practical requirement rather than a design preference: the regulation agent's P95 latency of 13.7 s is operationally acceptable precisely because it activates only when a regulatory check is warranted.

Future work follows four directions. First, a systematic retrospective protocol recording the orchestrator recommendation alongside the actual team decision on every strategically relevant lap across a full 24-race season would produce a quantitative concordance metric over the four discrete actions, replacing the current qualitative case-study approach with a season-level external validity measure. Second, expanding the radio corpus to approximately 2,000 examples under a multi-annotator protocol with inter-annotator agreement measurement is the immediate path to NER F_1 above 0.60; self-training over the existing 530 Whisper-transcribed messages could reduce the manual annotation burden. Third, replacing the manual if-else router with a mixture-of-agents model fine-tuned on orchestrator reasoning traces [41] could reduce the regulation-agent P95 tail and improve action consistency across edge cases. Fourth, replacing the parquet replay engine with a WebSocket consumer over the OpenF1 live API would verify that all six agents maintain acceptable latency under real-time data load, completing the transition from post-race analysis to live deployment.

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